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Simulation model for natural gas transmission pipeline network system

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Abstract

This paper focuses on developing a simulation model for the analysis of transmission pipeline network system (TPNS) with detailed characteristics of compressor stations. Compressor station is the key element in the TPNS since it provides energy to keep the gas moving. The simulation model is used to create a system that simulates TPNS with different configurations to get pressure and flow parameters. The mathematical formulations for the TPNS simulation were derived from the principles of flow of fluid through pipe, mass balance and compressor characteristics. In order to determine the unknown pressure and flow parameters, a visual C++ code was developed based on Newton-Raphson solution technique. Using the parameters obtained, the model evaluates the energy consumption for various configurations in order to guide for the selection of optimal TPNS. Results from the evaluations of the model with the existing TPNS and comparison with the existing approaches showed that the developed simulation model enabled to determine the operational parameters with less than ten iterations. Hence, the simulation model could assist in decisions regarding the design and operations of the TPNS.

Keywords: Transmission pipeline network; Compressor station; Simulation; Mathematical model; Energy;

1. Introduction

Natural gas is becoming one of the most widely used sources of energy in the world due to its environmental friendly characteristics. Usually, the location of natural gas resources and the place where the gas is needed for various applications are far apart. As a result, the gas has to be moved from deposit and production sites to consumers either by trucks in the form of liquefied natural gas (LNG) or through pipeline network systems. As reported in [1], short distances gas transportation by pipelines is more economical than LNG transportation. The LNG transportation incurs liquefaction costs irrespective of the distance over which it is moved. As a result, the development of transmission pipeline network system (TPNS) for natural gas is a key issue in order to satisfy the ever growing demand from the various customers [2].

When the gas moves by using the TPNS, the gas flows through pipes and various devices such as regulators, valves, and compressors. The pressure of the gas is reduced mainly due to friction with the wall of the pipe and heat transfer between the gas and the surroundings.

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Nomenclature

A_E, A_H	Constants for compressors equation
B_E, B_H	Constants for compressors equation
C_E, C_H	Constants for compressors equation
CS	Compressor station
D	Diameter
D_E, D_H	Constants for compressors equation
D_i	Customer located at I
D_{ij}	Diameter of pipe joining node i and j
D_l	Volumetric flow rate of load pipe l
E	Pipeline efficiency
f	Darcy's friction factor
G	Gas gravity
H	Adiabatic head
HP	Compression power
k	Specific heat ratio
K_{ij}	Pipe flow resistance
L	Length
L_{ij}	Length of pipe joining node i and j
LNG	Liquefied natural gas
M	Mass flow rate
$MMSCMD$	Million metric standard cubic meters per day
$MMSCFD$	Million standard cubic feet per day
n	Rotational speed of the compressor
n_c	Number of compressors
n_j	Number of junctions
n_l	Number of loops
n_p	Number of pipes
N_P	Number of unknown pressure variables
N_Q	Number of unknown flow variables
n_s	Number of compressor stations
N_{Total}	Total number of unknown variables
P	Pressure
P_s, P_d	Suction, discharge pressure
P_n	Standard pressure condition
q_j	Volumetric flow rate of outgoing pipe j
Q	Volumetric flow rate
Q_{Ci}	Gas flow rate to customer i
Q_i	Volumetric flow rate through pipe i
Q_n	Volumetric flow at standard conditions
R	Gas constant
t	Number of incoming pipe to a node
T	Temperature of gas
T_S	Suction temperature
T_n	Standard temperature conditions
$TPNS$	Transmission pipeline network system
u	Number of outgoing pipes from a node
w	Number of load pipes from a node
X	Vector representing unknown variables
Z	Gas compressibility
Z_1, Z_2	Suction, discharge side compressibility

Subscripts

d	discharge side of the compressor
E	Efficiency
H	head
i, j	upstream node, downstream node
s	suction side of the compressor

Greek letters

η_a	Adiabatic efficiency
γ	The ratio of specific heat

Compressor stations are usually installed to boost the pressure of the gas and keep the gas moving to the required destinations. It is estimated that 3 to 5% of the gas transported is consumed by the compressors in order to compensate for the lost pressure of the gas [3, 4]. This is actually a huge amount of gas especially for the network transmitting large volume of gas. At the current price, this represents a significant amount of cost for the nation operating large pipeline network system. For instance, considering the U.S. TPNS, Wu [5] indicated that a 1% improvement on the performance of the transmission pipeline network system could result a

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saving of 48.6 million dollars. Carter [6] also presented that the cost of natural gas burned to power the transportation of the remaining gas for the year 1998 is equivalent to roughly 2 billion dollars for U.S. transmission system. Investigation on various TPNS indicated that the overall operating cost of the system is highly dependent upon the operating cost of the compressor stations which represents between 25% and 50% of the total company's operating budget [3, 7]. Hence, compressor station is considered as one of the basic elements in TPNS.

The main issues associated with both design and operating TPNS are minimizing the energy consumption and maximizing the flow rate through pipes. Over the years, numerical simulations of TPNS have been carried out in order to determine the optimal operational parameters for given networks with various degrees of success [4, 8-12]. From the optimization perspective, the problem of developing an optimal TPNS is nonlinear programming problem where the objective function is typically nonlinear and non-convex, and some of the constraints are also nonlinear. Different techniques were proposed in order to get the optimal parameters of TPNS by either modifying the objective function or relaxing some of the constraints [11, 13-16]. However, due to the complexity of the objective function and the constraints, the determination of optimal parameters for TPNS is yet challenging from the optimization perspective.

On the other hand, simulation has contributed significant achievements in analyzing the TPNS problems [17-22]. TPNS simulation is used to determine the design and operating variables of the pipeline network for various configurations.

The complexity of the simulation analysis depends on the extent of the pipeline network configurations (gunbarrel, branched, looped, etc.), the nature of the gas (single phase dry gas, two-phase gas-liquid mixture) and other factors such as temperature of the gas, the number of sources of the gas (single source, multi-source) and internal pipe corrosion.

TPNS consists of pipes and non-pipe elements such as compressors, regulators, valves, scrubbers, etc. The simulation of TPNS system without the non-pipe elements is relatively easier to handle and developed by Osiadacz [17]. The addition of non-pipe elements makes the simulation of TPNS more complex due to the modeling of the non-pipe elements. More equations have to be added into the governing simulation equations when the non-pipe elements are considered during analysis. Compressor station is one of the main non-pipe components of gas transmission system and considered as a key element.

One of the basic differences among TPNS simulation analysis models with non-pipe elements is in the way compressor station is modeled during simulation. There have been attempts reported by various researchers on modeling compressor stations within the TPNS during simulation. The various compressor simulation models including the compressor instability are summarized in [23]. One of the options in compressor station modeling is to consider the compressor station as a black box by setting either the suction or discharge pressures [24]. Only little information can be obtained to be incorporated into the simulation model to represent the compressor station. The effect of compressor station during simulation of TPNS has been incorporated by pre-setting the discharge pressures [17, 25]. However, the speed of the compressor, suction pressure, suction temperature, and flow through the compressor were neglected during the analysis. Even though there have been attempts reported regarding the simulation of TPNS with non-pipe elements, there are issues that are not fully addressed.

The main objective of this paper is to develop a TPNS simulation model for the analysis of the performance of pipeline network system incorporating compressor characteristics, effect of two-phase flow and the age of the pipes.

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The developed simulation model focused on determining the nodal pressure and flow parameters which are essential to evaluate the performance of the TPNS. It incorporates the detailed characteristics of the compressor stations. The flow equation in the simulation model has been made flexible to include the effect of multiphase flow and the effect of corrosion of the pipes. The Newton-Raphson based solution procedures were implemented using visual C++. Various configurations of the TPNS could be generated and evaluated using the simulation model to identify the system with minimum energy consumption. As a result, the simulation model could assist in decisions regarding the design and operations of the TPNS. The paper continues in section 2 with detailed mathematical formulations of the basic governing simulation equations. Section 3 shows description of Newton-Raphson based solution procedures for the simulation model. The numerical evaluation of the simulation model based on the existing pipeline network system is presented in section 4. The comparison of the simulation model with the previous approach is discussed in section 5. Finally, conclusion of the study and directions for future research are presented in section 6.

2. Mathematical formulation for the simulation model

The mathematical model for the TPNS simulation is developed based on the knowledge of the performance characteristics of the compressors, equations which govern the flow of the gas through pipes, and the principles of conservation of mass. The mathematical formulation and the types of equations incorporated into the governing simulation equations depend on the configurations of the network, the nature of the gas, and internal corrosion. Fig. 1 shows the general procedure of the mathematical formulation based on configurations and basic elements of TPNS to form the governing simulation equations.

< Insert Figure 1 here>

Fig. 1. General procedure for mathematical formulation of TPNS simulation.

2.1 Formulation of pipe flow equations

Pipe flow equation is one of the governing equations for the simulation. It is derived based on the principle of the flow analysis of gas in pipes. The flow of gas through pipes can be affected by various factors such as gas properties, friction factor and the geometry of the pipes. The relationship between the upstream pressure, downstream pressure and flow of the gas in pipes can be described by various equations[17, 18]. For this study, general flow equation is used due to its frequent application in gas industry to describe the relationship between pressure difference and gas flow in pipes.

Single phase flow equation for a pipeline element (Fig. 2) relating upstream pressure P_i, downstream pressure P_j and the flow through pipe Q_{ij} can be expressed as:

$$P_i^2 - P_j^2 = K_{ij} Q_{ij}^2 \tag{1}$$

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where K_{ij} takes different forms depending on the dimensions of the parameters used in the flow equations. For instance, when $P[kPa]$, $T[K]$, $L[km]$, $Q[m^3/hr]$ and $D[mm]$, the expression for K_{ij} takes the form:

$$K_{ij} = 4.3599\times10^8 \frac{fGZT}{D^5}\left(\frac{P_n}{T_n}\right)^2 L \tag{2}$$

Eq. (1) can be represented as functional form, where the representation consists of only parameters which are unknown in that equation. If all nodal pressures and flow rate are unknown, the functional representation takes the form as

$$F(P_i, P_j, Q_{ij}) = 0 \tag{3}$$

< Insert Figure 2 here>

Fig. 2. Pipe joining two consecutive nodes.

Single phase flow modeling approaches may not be adequate to predict the transport capabilities of the pipelines required to move fluids mixtures. As suggested in [26-28] , a two-phase flow analysis or single phase flow analysis with modified friction factor may be required to adequately predict the transport capabilities of such system. Furthermore, corrosion in oil and gas industries is one of the serious challenges which affect the performance of TPNS. As the age of the pipe increases the roughness of the pipe will tend to increase due to the accumulation of various elements around the internal surface of the pipe. Therefore, Eq. (1) should have to be modified to take into consideration the effect of multiphase flow and corrosion in pipes.

2.2 The looping conditions

In order to reduce the pressure drop in a certain section of the pipeline due to pressure limitation or for increasing the flow rate in bottleneck sections, looped pipe network may be constructed. Looped piping system, shown in Fig. 3, consists of two or more pipes connected in such a way that the gas flow splits among the branch pipes and eventually combine downstream into a single pipe. When the TPNS contains loops, additional equations must be incorporated to the flow equations. These additional equations are obtained from looping condition. The looping condition states that for each closed loop within the network system the pressure drop is zero[17, 18].

< Insert Figure 3 here>

Fig. 3. Part of pipeline network with loop.

Based on the looping condition, for the TPNS shown in Fig. 3, the pressure drop in pipe branch 1-2-4 must equal the pressure drop in pipe branch 1-3-4. This is due to the fact that both pipe branches have a common starting point (node 1) and common ending point (node 4). The

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looping condition for the pipeline network shown in Fig. 3 can be expressed based on the single phase general flow equation as:

$$\frac{Q_2}{Q_3} = \left(\frac{L_2}{L_1}\right)^{0.5} \left(\frac{D_1}{D_2}\right)^{2.5} \tag{4}$$

where Q_2, L_1, and D_1 are the flow rate through pipe, the length and the diameter for pipe 1-2-4, respectively. Similarly, Q_3, L_2, and D_2 are the flow rate through pipe, the length and the diameter for pipe 1-3-4, respectively.

2.3 Formulation of compressor equations

Usually, the data related to compressor are available in the form of compressor performance characteristics map. In order to integrate the characteristics map of the compressor into the simulation model, it is necessary to approximate the characteristics map with mathematical models. The basic quantities related to a centrifugal compressor unit are inlet volume flow rate Q, speed n, adiabatic head H, and adiabatic efficiency η. The mathematical approximation of the performance map of the compressor can be done based on the normalized characteristics. The three normalized parameters which are necessary to describe the performance map of the compressor includes, H/n^2, Q/n and η [29]. Based on the normalized parameters, the characteristics of the compressor can be approximated either by two degree [30] or three degree polynomials [3]. Three degree polynomial which gives more accurate approximation is used in this paper. Applying the principles of polynomial curve-fitting procedures for each compressor, the relationship among the basic normalized parameters can be best described by the following two equations:

$$\frac{H}{n^2} = A_H + B_H\left(\frac{Q}{n}\right) + C_H\left(\frac{Q}{n}\right)^2 + D_H\left(\frac{Q}{n}\right)^3 \tag{5}$$

$$\eta = A_E + B_E\left(\frac{Q}{n}\right) + C_E\left(\frac{Q}{n}\right)^2 + D_E\left(\frac{Q}{n}\right)^3 \tag{6}$$

where, A_H, B_H, C_H, D_H , A_E, B_E, C_E, D_E are constants which depend on the unit. A set of data of the quantities Q, n, H, and η can be collected by testing the unit and the constants could be determined by the application of the method of least squares technique.

In considering the effect of compressors for the TPNS simulation model, the relationships as in equation (5) and equation (6) might not be used directly. The information from the compressor map should have to relate the discharge pressure, the suction pressure and flow rate. The relationships between suction pressure P_s, and discharge pressure P_d with the head H is given as [29] :

$$H = \frac{ZRT_S}{m}\left\{\left[\frac{P_d}{P_s}\right]^m - 1\right\} \tag{7}$$

where $m = (k-1)/k$ with k to be specific heat ratio, R is gas constant, T_S is the suction temperature and Z is the compressibility of the gas.

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Substituting the value of H from equation (5) into equation (7) and rearranging yields the required compressor performance equation which can be incorporated as governing equation for the simulation model.

$$\left(\frac{P_d}{P_s}\right)^m = \frac{mn^2}{ZRT_S}\left\{A_H + B_H(Q/n) + C_H(Q/n)^2 + D_H(Q/n)^3\right\} + 1 \tag{8}$$

Equation (8) represents a general compressor equation for single compressor. It can be seen that most of the parameters that describe the compressor are incorporated in the general compressor equation. This is one of the significant contributions in the area of simulation of TPNS with compressor stations as non-pipe elements. Fig. 4 shows a comparison of the plot of the performance characteristics of the compressor generated by Eq. (5) and actual data collected from the performance map of the compressor in [31].

< Insert Figure 4 here>

Fig. 4. Comparison of selected data and approximated data for typical centrifugal compressor.

Eq. (8) can also be represented with short functional form as $f(P_D, P_S, Q) = 0$ if all the suction side pressure, discharge side pressures and the flow rates are unknown.

For compressors operating in parallel within the stations, the general compressor equation can be modified to take into account the number of compressors working within the station as:

$$\left[\frac{P_d}{P_s}\right]^m = \frac{m\ n^2}{ZRT_S}[A_H + B_H\left(\frac{(Q/e)}{n}\right) + C_H\left(\frac{(Q/e)}{n}\right)^2 + D_H\left(\frac{(Q/e)}{n}\right)^3] + 1 \tag{9}$$

where e is the number of compressors working in parallel within the compressor station.

2.4 Formulation of mass balance equations

In addition to the pipe flow and compressor equations discussed above, mass balance provides the remaining basic equations in order to have a complete mathematical formulation for the simulation of a given TPNS.

< Insert Figure 5 here>

Fig. 5. Mass balance formulation at junction c of a TPNS.

The mass balance equations are obtained based the principle of conservation of mass at each junction of TPNS. At any junction c within a TPNS, Fig. 5, the generalized mass balance equation for t incoming pipes, u outgoing pipes and w load pipes can be summarized as:

$$\sum_{i=1}^{i=t} Q_i - \sum_{j=1}^{j=u} q_j - \sum_{l=1}^{l=w} D_l = 0 \tag{10}$$

where $Q_1, Q_2, ..., Q_t$ are flow through incoming pipes to junction c, $q_1, q_2, ..., q_u$ are flow through outgoing pipes from junction c and $D_1, D_2, ..., D_l$ are the load from junction node c.

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If all the flow rates through the incoming and outgoing pipes are unknown. The functional representation of Eq. (10) takes the form:

$$f(Q_1, Q_2, ..., Q_t,\ q_1, q_2, ..., q_u) = 0 \tag{11}$$

3. Solution Procedures for the Simulation Model

A TPNS configuration with n_p number of pipes, n_c number of compressor stations with compressors working in parallel, n_l number of loops, and n_j number of junctions is assumed for the analysis. The number of unknown nodal pressures to be determined would be $[(n_p + n_c) - (n_j + 1)]$ and that of unknown flow parameters would be $[2n_l + 2n_j + 1]$. On the other hand, there are n_p pipe flow equations, n_c compressor equations, $2n_l$ equations from looping and n_j mass balance equations. As a result, $(n_p + n_c + 2n_l + n_j)$ equations with $(n_p + n_c + 2n_l + n_j)$ unknowns make the TPNS problem solvable.

After the basic equations for TPNS are developed, the results for unknown parameters were determined on the basis of Newton-Raphson technique. Newton-Raphson technique is powerful for analysis of TPNS problems with large number of unknown parameters [32-35].

Let $N_P =$ total number of unknown pressure parameters.

$N_Q =$ total number of unknown flow parameters.

The total number of unknown parameters N_{Total} is given as:

$$N_{Total} = N_P + N_Q \tag{12}$$

The set of pipe flow, compressor, mass balance, and looping equations can be represented as

$$\left\{ \begin{array}{l} F_1(P_1, P_2, \ \bullet\bullet\bullet, P_{N_P}, Q_1, Q_2, \ \bullet\bullet\bullet, Q_{N_Q}) = 0 \\ F_1(P_1, P_2, \ \bullet\bullet\bullet, P_{N_P}, Q_1, Q_2, \ \bullet\bullet\bullet, Q_{N_Q}) = 0 \\ \quad \bullet \\ \quad \bullet \\ \quad \bullet \\ F_{N_{Total}}(P_1, P_2, \ \bullet\bullet\bullet, P_{N_P}, Q_1, Q_2, \ \bullet\bullet\bullet, Q_{N_Q}) = 0 \end{array} \right\} \tag{13}$$

Eq.(13) can be written in matrix form as in [36]

$$\tilde{F}\left(\tilde{X}\right) = \tilde{0} \tag{14}$$

where the vector X represents the total number of unknown pressure and flow parameters.

The multivariable Newton-Raphson iterative procedure for Eq. (14) takes the form

$$\tilde{X}_{new} = \tilde{X}_{old} - \left[\tilde{A} \Big|_{\tilde{X}_{old}} \right]^{-1} \tilde{F}\left(\tilde{X}_{old}\right) \tag{15}$$

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where $\tilde{A}$ is the Jacobian matrix whose elements are partial derivatives of the functions with respect to the unknown pressure and flow parameters.

The matrix $\tilde{A}$ in Eq. (15) is defined as

$$\tilde{A} = \begin{bmatrix} \frac{\partial F_1}{\partial P_1} & \cdots & \frac{\partial F_1}{\partial P_{NP}} & \frac{\partial F_1}{\partial Q_1} & \cdots & \frac{\partial F_1}{\partial Q_{N_Q}} \\ \frac{\partial F_2}{\partial P_1} & \cdots & \frac{\partial F_2}{\partial P_{NP}} & \frac{\partial F_2}{\partial Q_1} & \cdots & \frac{\partial F_2}{\partial Q_{N_Q}} \\ \bullet & & & & & \\ \bullet & & & & & \\ \bullet & & & & & \\ \frac{\partial F_{NTotal}}{\partial P_1} & \cdots & \frac{\partial F_{NTotal}}{\partial P_{N_P}} & \frac{\partial F_{NTotal}}{\partial Q_1} & \cdots & \frac{\partial F_{N_{Total}}}{\partial Q_{N_Q}} \end{bmatrix} \tag{16}$$

From Eq. (16), the inverse of the Jacobian matrix should be computed for each iteration. However, there is another approach that does not require the rigorous computation of associated with the inversion of the Jacobian matrix. Eq. (16) can be rewritten as

$$\tilde{A}\Big|_{X_{old}} \left[\tilde{X}_{new} - \tilde{X}_{old} \right] = -\ \tilde{F}\left(\tilde{X}_{old} \right) \tag{17}$$

The value of the unknown parameters can be calculated from Eq. (17) iteratively until the relative errors are less than specified tolerance or the number of iterations equal to the desired value. Fig. 6 shows the flowchart of the general simulation model. A visual C++ code was developed for the simulation based on Newton-Raphson solution technique. For efficient operation of the simulation model, the overall code is grouped in to several subtasks. These include, subtask for mathematical formulation, subtasks for matrix elements generation, subtask for input data, subtask for Gaussian eliminations, subtask for error sorting and subtask for evaluating the networks. The solution procedures were converged to final solution depending on the initial estimation. Usually, the values of the unknown pressure and flow parameters were obtained with less than ten iterations.

The simulation model was tested by giving wide range of initial estimations for the unknown parameters and convergence was achieved in most of the attempts. The user might obtain the final solution easily by varying the initial estimations. For instance, proper estimation for the nodal pressures could be obtained from the exit pressure requirements at the various demand stations.

The developed simulation model enabled to create and analyze various TPNS configurations. The model could also facilitate for the users to analyze and compare the various TPNS configurations in order to develop a network with minimized energy consumption. The analysis can be done by varying the diameters of the pipes used in the network, the number of compressor stations working on the network, the speed of the compressors, and length of pipeline. The

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different TPNS configurations can then be compared on the basis of energy consumption to select an optimal TPNS which meets the demand requirements.

< Insert Figure 6 here>

Fig. 6. Flow chart of the simulation model based on Newton-Raphson technique.

4. Application of the simulation model

The simulation model was tested based on the data taken from part of the existing pipeline network system. The TPNS in Fig. 7 shows part of the existing network which transmits gas from source to five different power generating plants with various capacities. The pressure and flow requirements are different for all power plants. Various scenario analyses were performed using the simulation models which are very common to the existing TPNS. Because of the limited access to the compressor working in the field, a centrifugal compressor data from the manufacturer was used for the simulation model. The network consists of all the three most common configurations of pipeline network systems, i.e. branched, gunbarrel (linear) and looped configuration.

4.1 Solution to nodal pressure and flow variables using TPNS simulation model

In the numerical evaluations, the following data were used throughout the analysis. All the properties of the gas were collected from the nearest operational gas transmission company. For the compressor station, all units within the stations are assumed to be identical and are connected in parallel. The data related to pipe diameters, lengths and customer requirements are based on existing TPNS and literatures. The gas is a mixture of methane (92 %), ethane (5%), nitrogen (1%) and others (2%). Gas gravity $G = 0.5$, the average gas flowing temperature $T = 308K$, base pressure $P_n = 101kPa$ base temperature $T_n = 288K$, gas constant for air $R_{air} = 287.5J/kg\,K$, gas compressibility factor $Z = 0.91$, isentropic exponent $k = 1.287$ and ten years service of pipes are used for the analysis. The dimensions for the parameters are $P[kPa]$,$T[K]$, $L[km]$, $Q[m^3/hr]$ and $D[mm]$. The pipe flow resistance $K_{ij} = 6.4575E + 07 \times L/D^5$. For the compressor station, all units within the stations are assumed to be identical. The coefficients for the approximation of the performance map of the compressor are determined to be: $A_H = 6.35184E-05$, $B_H = -7.08347E-05$, $C_H = 2.54105E-05$, $D_H = -2.92486E-06$.

< Insert Figure 7 here>

Fig. 7. Part of the existing natural gas transmission pipeline network system.

The TPNS in Fig. 7 consists of 10 pipes, 1 compressor station, 1 loop and 4 junctions. As a result, there are 6 nodal pressures and 11 flow parameters to be determined. Therefore a total of 17 independent equations should have to be obtained in order to solve the network problem. For

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this, there are 10 pipe flow equations, 1 compressor equation, 2 looping equations and 4 mass balance equations which form the 17 independent equations to analyze the TPNS.

The network was analyzed with the aid of the developed simulation model under three conditions which include single phase gas flow analysis, two phase gas-liquid flow analysis and single phase flow with corrosion. The results for the unknown pressure and flow parameters were obtained with less than ten iterations. The maximum relative percentage error at the end of the 10th iteration was less than 10^{-11}. The simulation model was tested by varying the demand requirements, number of compressors working within the station, compressor speed, pipe diameter, and source pressure.

Fig. 8 shows the convergence of nodal pressures to the final pressure solution for the first ten iterations at node 1 and 2 of the TPNS with source pressure of 2500kPa to meet a customer requirement of 4000kPa. The convergences of the remaining nodal pressures follow the same trend as that of the pressure at node 2. The convergence of the corresponding flow parameters are shown in Fig. 9 and Fig. 10. From the convergence graphs of both the pressure and flow variables, it is observed that it took only from 4 to 6 iterations for the solution to get stable.

< Insert Figure 8 here>

Fig. 8. Convergence of nodal pressures for single phase gas flow.

< Insert Figure 9 here>

Fig. 9. Convergence of the flow parameters through main pipes for single phase gas flow.

< Insert Figure 10 here>

Fig. 10. Convergence of flow parameters through branch pipes for single phase gas flow.

4.2 Comparison of nodal pressures and flow variables based on different flow conditions

The developed TPNS simulation model can be used to compare and evaluate different flow conditions. Comparison was made between the three flow analysis .i.e. two phase gas-liquid flow analysis with 0.005 liquid holdup, single phase flow with corrosion effect and single phase flow with no corrosion effect. For the case of single phase with corrosion effect, it is assumed that the pipes were served ten years. The comparison showed that the existence of 0.005 liquid holdup had a significant effect on both pressure and flow parameters. Note that 0.005 liquid holdup is the maximum amount that might exist in gas transmission pipelines [37]. In order to meet the same requirement, two phase flow analysis required high compression ratio with reduced capacity of the pipes compared to single phase flow analysis with or without corrosion effect. Comparison of single phase with and with no corrosion effect had also been done. For ten years service life of pipe, the effect of corrosion is not that much significant when the TPNS was modeled by neglecting corrosion. This could be resulted because of the internal coating of the pipes that reduce the rate of corrosion. However, the effect of corrosion increased when the service life of the pipe increased. Fig. 11 shows the comparison of nodal pressures for single phase gas flow, two phase gas-liquid flow, and single phase with corrosion effect. The corresponding comparison of the flow parameters is shown in Fig. 12.

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< Insert Figure 11 here>

Fig. 11. Comparison of nodal pressures.

< Insert Figure 12 here>

Fig. 12. Comparison of flow through pipes.

4.3 Evaluation of power consumption for the given TPNS

After the pressure and flow parameters are obtained, the simulation model evaluates the energy consumption of the TPNS for various configurations in order to guide for the selection of optimal system. In this paper, energy consumption by system was considered to evaluate alternative networks. The different alternatives are then compared based on Eq. (18) to select the alternative with minimized energy consumption.

As suggested in [18], the amount of energy input to the gas by the compressors is dependent upon the pressure of the gas and flow rate. The power required by the compressor that takes into account the compressibility of gas is given by

$$HP = 4.0639\left(\frac{\gamma}{\gamma-1}\right)QT_1\left(\frac{Z_1+Z_2}{2}\right)\left(\frac{1}{\eta_a}\right)\left[\left(\frac{P_2}{P_1}\right)^{\frac{\gamma}{\gamma-1}}-1\right] \tag{18}$$

where HP is the compression power in horsepower , γ is the ratio of specific heats of gas , Q is gas flow rate, T_1 is suction temperature of the gas, P_1 is suction pressure of the gas, P_2 is discharge pressure of the gas, Z_1 is compressibility of the gas at suction condition, Z_2 is compressibility of the gas at discharge condition, η_a is compressor adiabatic (isentropic) efficiency.

4.4 Analysis using TPNS simulation model

Various analyses were performed using the simulation model for the TPNS shown in Fig. 7. The influences of the addition of new customers to the system and departure of the existing customers from the system on the performance of the TPNS were analyzed. Furthermore, the performance of the compressor stations for various demand pressure requirements was analyzed with the developed simulation model.

Fig. 13 shows the effect of change in exit pressures at the demand stations on the main flow (Q_1) for various speeds. The corresponding variations of the compression ratio (CR) of the compressor are also shown on the figure.

< Insert Figure 13 here>

Fig. 13. Variation of the exit pressure at the demand station on main flow.

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5. TPNS model validation

The developed TPNS simulation model provided the required operational variables of the pipeline network for various configurations. The results from the simulation model needs simulation verification and validation for accuracy and reliability [38-41]. The techniques used in order to validate the TPNS simulation model were comparison with the previous models. Two cases were taken from the available literatures in order to validate the results of the developed TPNS simulation model.

5.1 Comparison based on gunbarrel pipeline network

The results of TPNS simulation model for the gunbarrel network configurations are compared with the method proposed by Wu et al [3]. The pressure range for each node is between 689.5 kPa and 6895 kPa. There are 5 compressors within each compressor stations. The flow through pipes is assumed to be 600 million standard cubic feet per day (MMSCFD) or equivalent to 707,921 million metric standard cubic meters per day (MMSCMD). The problem for the network was the determination of nodal pressures which minimizes the fuel cost of the compressors.

Wu et al. [3] determined solutions for the optimal nodal pressures for minimizing the fuel consumption of the system using exhaustive search method.

The pipeline network was also analyzed using the TPNS simulation model. In TPNS simulation model, it is assumed that maintaining pressure requirement was critical and therefore, the source pressure and the demand pressure were assumed to be known. Therefore, the problem in TPNS simulation is finding the remaining nodal pressures, flow rate, number of compressors within the stations, power consumption and the speed of the compressors which satisfies the requirement.

It was observed that as the speed of the compressors increased, the results of nodal pressures from the TPNS simulation were getting closer to the results of nodal pressures obtained by Wu et al [3]. However, an increase in deviation of flow parameter was observed when speed of compressors increased. After conducting simulation experiments, compressor speed of 5775 rpm gave better results of nodal pressures with maximum percentage error of 2.37% which was obtained at node 5 of the network. The corresponding flow deviation was 10.71%. Fig. 14 shows comparison of the result of nodal pressures obtained from the TPNS simulation model at speed of 5775 rpm and the results obtained in [3].

< Insert Figure 14 here>

Fig. 14. Comparison of nodal pressures for gunbarrel TPNS

Wu et al [3] obtained the solution for nodal pressures so that the fuel consumption for the system is minimized. The results obtained from TPNS simulation model shows that the simulation model could be used to compare various operations of the compressor. This could help to compare the alternatives and select the one with minimum power consumption.

5.2 Comparison based on looped pipeline network

The model was also compared with another method based on the problem instance taken from [17]. Fig. 15 shows the TPNS considered for the comparison. The author in [17] applied a Newton loop – node method in order to get the flow and pressure parameters. Furthermore, the author assumed fixed compression ratios of 1.4 and 1.8 for compressor station 1(CS1) and

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compressor station 2 (CS2), respectively. However, the final nodal pressures obtained after achieving the predefined error requirements failed to satisfy the previously assumed compression ratios. The compression ratios were actually 1.34 and 1.18 for CS1 and CS2, respectively.

< Insert Figure 15 here>

Fig. 15. TPNS with 10 pipes and two compressor stations[17].

The simulation experiments were conducted by varying the number of compressors and speed of compressors using the developed TPNS simulation model. The analysis was conducted using the performance characteristics of the compressors taken from [31]. Similar compressors were assumed to work on both compressor stations. From the simulation experiments, it was observed that one compressor for each compressor station is sufficient to meet the customer requirements.

Simulation experiments based on the given compressors characteristics showed that, the two compressor stations have to work nearly with the same compression ratios. An increase in compression ratio for the first station could result more flow through pipe 1(see Fig. 15) and reversal flow through pipe 3. This might cause flow reversal in the second compressor station. For instance, the compressor ratio mentioned for CS2 by Osiadacz [17] was achieved when the compressor runs with speed of 4800rpm. Based on this reference speed for CS1, the speed of the compressor increased to improve the compression ratio of CS2. The compressor speed at CS1 was increased till flow reversal starts to CS (i.e. 5450 rpm). It was observed that, the deviations of the nodal pressures and flow variables increased as the speed increase towards the speed of 5450 rpm.

After conducting simulation analyses based on the requirements, compressors speeds of 5025 rpm for CS1 and 4750 rpm for CS2 gave results of nodal pressures and flow variables close to Osiadacz [17]. Mean absolute percent error of 5.10% was observed between the two methods. The variations of the flow and nodal pressure variables could be as a result of the type of flow equations that were used in the analysis and the oversimplification of compressor stations in the case of the method in [17]. Panhandle 'A' flow equation was used in [17] where as general flow equation has been applied in the developed TPNS simulation. As developed in section 2, the TPNS simulation model consists of detailed characteristics of the compressor rather than only limited to compression ratio.

The results of comparison of the TPNS simulation model and that of Osiadacz [17] for flow rate through pipes is shown in Fig. 16. From the figure, it is observed that the maximum deviation between the results of TPNS simulation model and the results from Osiadacz [17] occurred at pipe 1, 4 and 6. This is as a result of the higher compression ratio assumed at CS1 which was far from the calculated value. The negative flow observed in pipe 3 of the network showed that the actual flow direction for the gas is opposite to the assumed direction. Hence, the actual flow in pipe 3 is from node 3 to node 2.

< Insert Figure 16 here>

Fig. 16. Comparison of flow rates between TPNS simulation and results of Osiadacz [17]

The comparison of the corresponding nodal pressures between the two methods is shown in Fig. 17. From the figure, it is observed that higher deviation at nodal pressures between the Osiadacz [17] and the results from the developed TPNS simulation model happened at node 4.

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This is as a result of the value of the CR assumed at CS1. Generally, the results of nodal pressure from the TPNS simulation model is higher than the nodal pressures obtained from Osiadacz [17]. This could be as a result of the flow equations used in both methods. The general flow equation which was used in TPNS simulation model result less pressure drop compared to the Panhandle 'A' flow equation which was used by Osiadacz [17].

< Insert Figure 17 here>

Fig. 17. Comparison of flow rates between TPNS simulation and results of Osiadacz [17]

6. Conclusion

In order to analyze the various configurations of natural gas pipeline transmission network system, a simulation model was developed by incorporating the detailed parameters of the compressors. Speed of the compressor, flow rate, suction pressure, discharge pressures and suction temperature are included in the general compressor equation. These are critical elements which affect the performance of the transmission system.

The developed TPNS simulation model enabled to determine pressure and flow parameters of the network under various conditions. The results for the unknown pressure and flow parameters were obtained with less than ten iterations. From the convergence graphs of both the pressure and flow variables, it is observed that it took only from 4 to 6 iterations for the solution to get stable. The maximum relative percentage error at the end of the 10th iteration was less than 10^{-11}. The performance of the system which includes, power consumption, compression ratio and flow capacity could be analyzed based on the parameters obtained. As a result, the model could assist in decisions regarding the design and operations of the pipeline network.

The results of the TPNS simulation model were compared with two other models based on various pipeline network configurations. In the first case, the simulation model was compared to an exhaustive optimization technique based on gunbarrel pipeline networks system. The model yielded close solutions of nodal pressures with less than 1.8% absolute parentage errors. The comparison based on looped pipeline network also showed that, the TPNS simulation model was able to provide solutions to nodal pressures and flow variables with mean absolute error of 5.10 % between the two methods.

The developed simulation model could be easily extended to be applied for the analysis of pipeline network systems for other petroleum products. When pipeline network operates, there are cases where the system may be subjected to various sever environmental factors like change in temperature and corrosion. The simulation model to take into account these effects during modeling is an important issue. Transient, or time dependent, model is an important problem to be addressed from the simulation perspective of gas transmission pipeline network systems.

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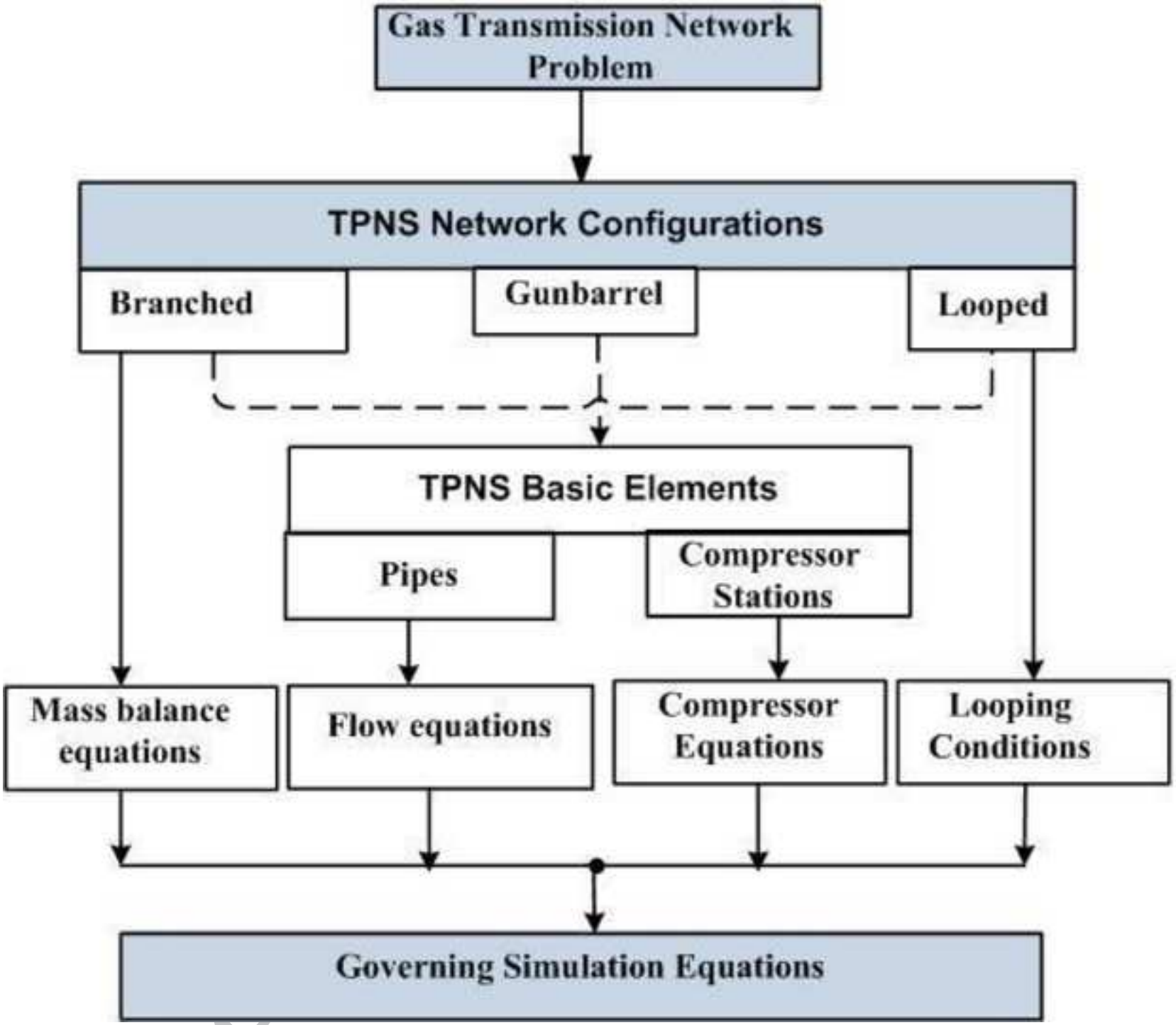

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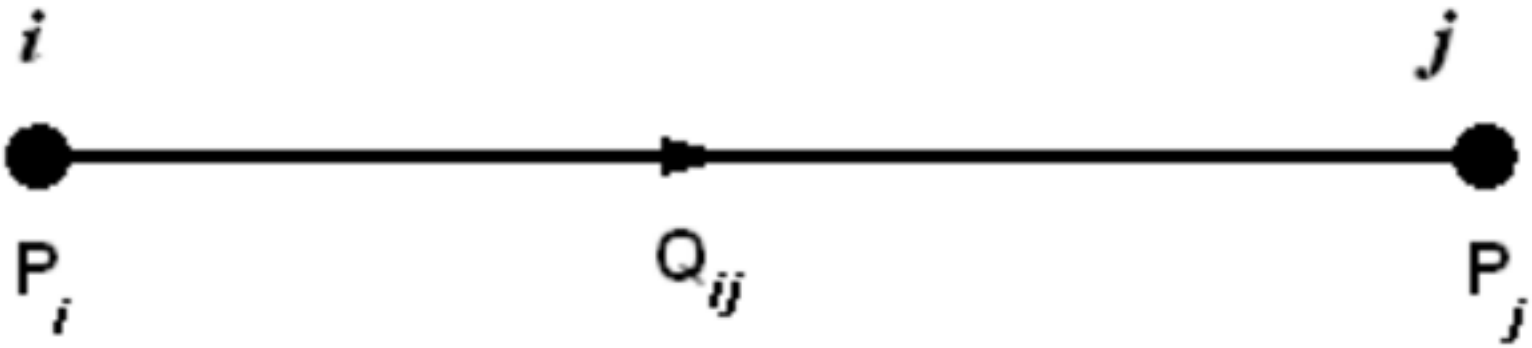

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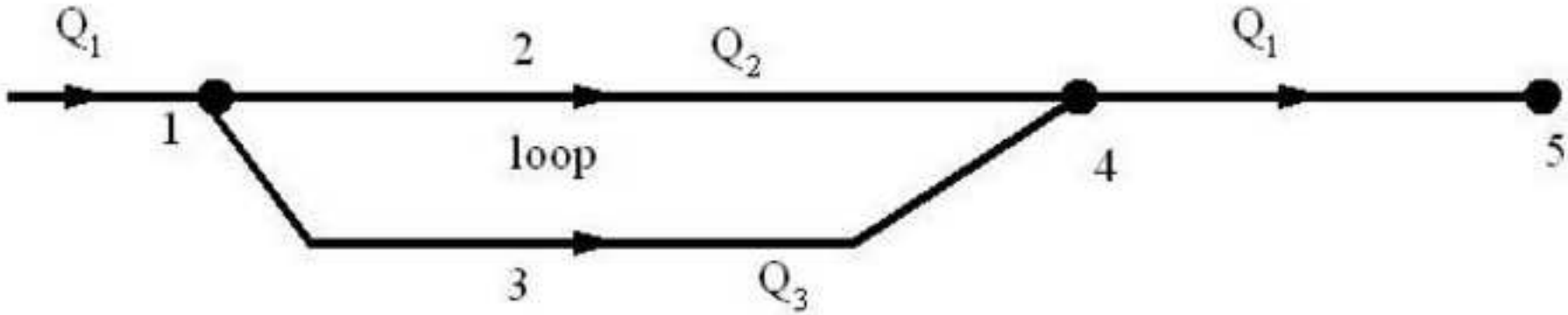

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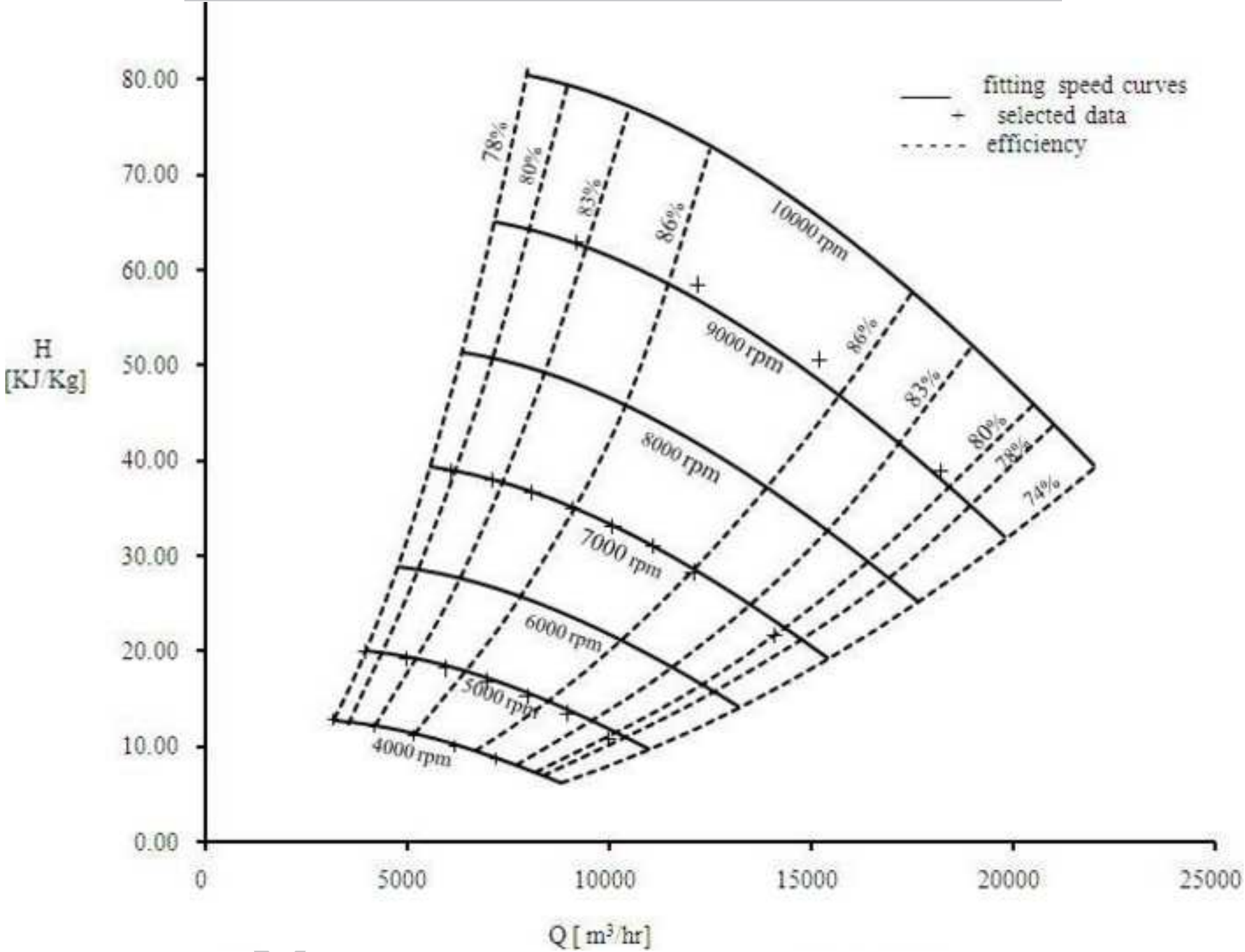

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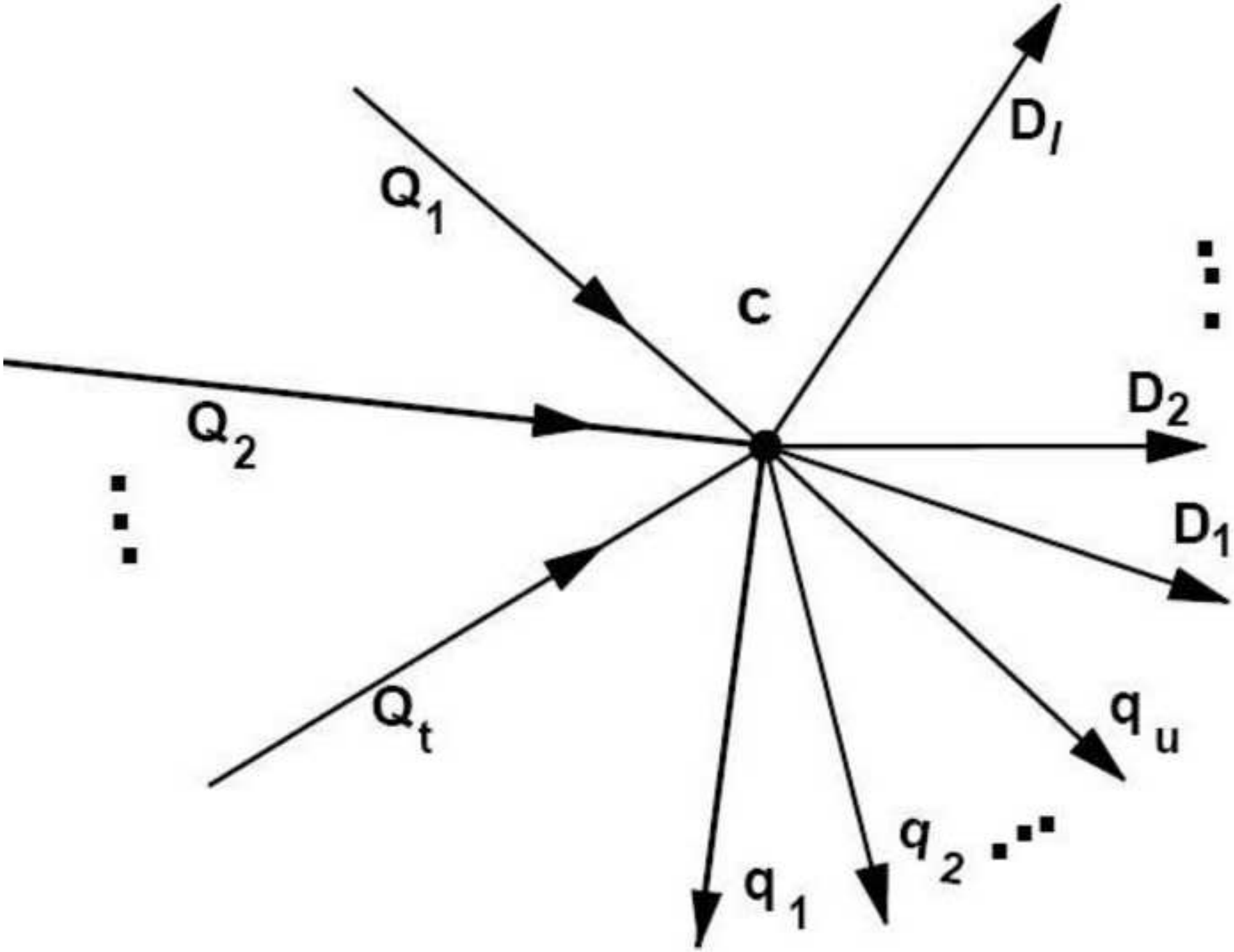

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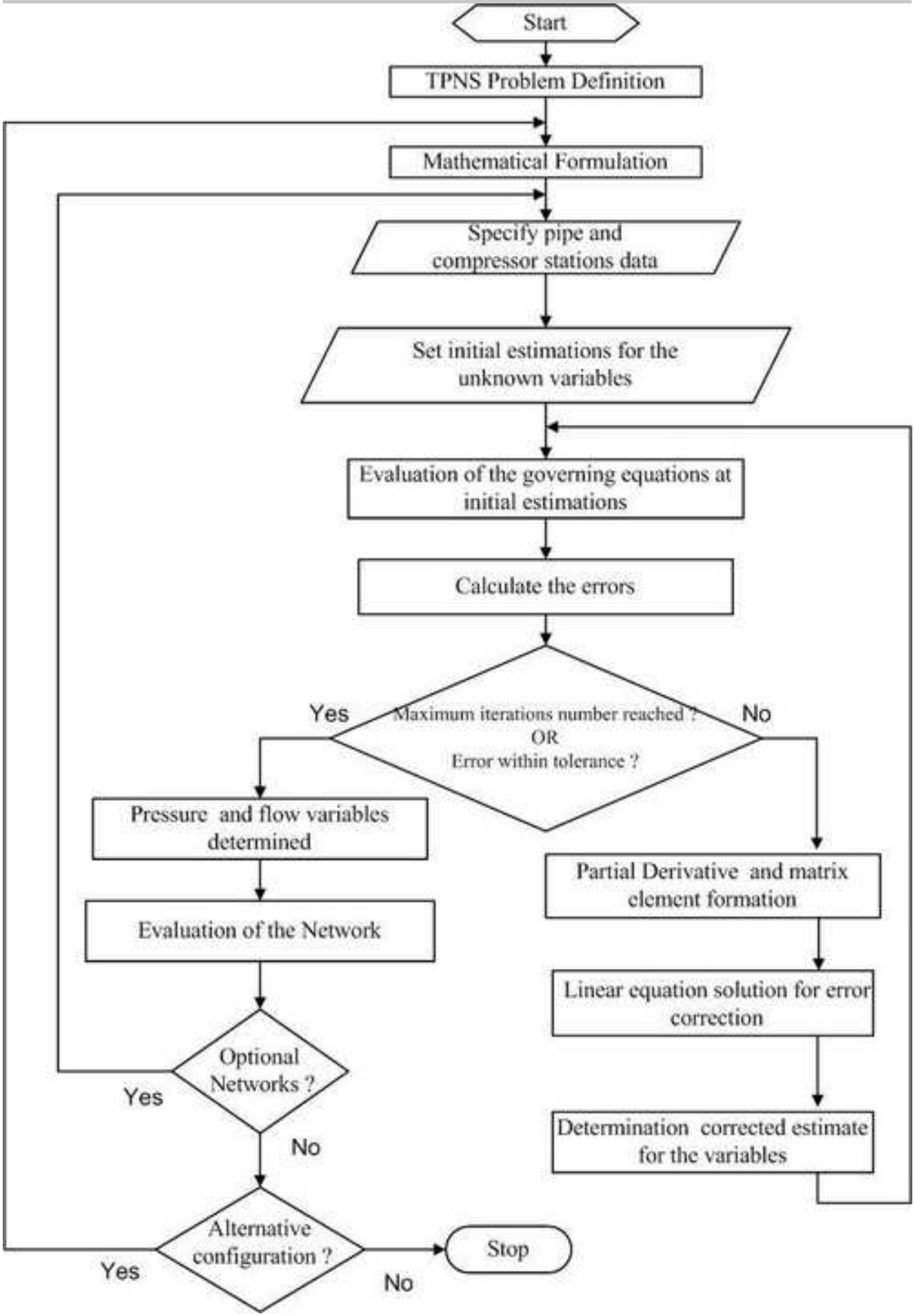

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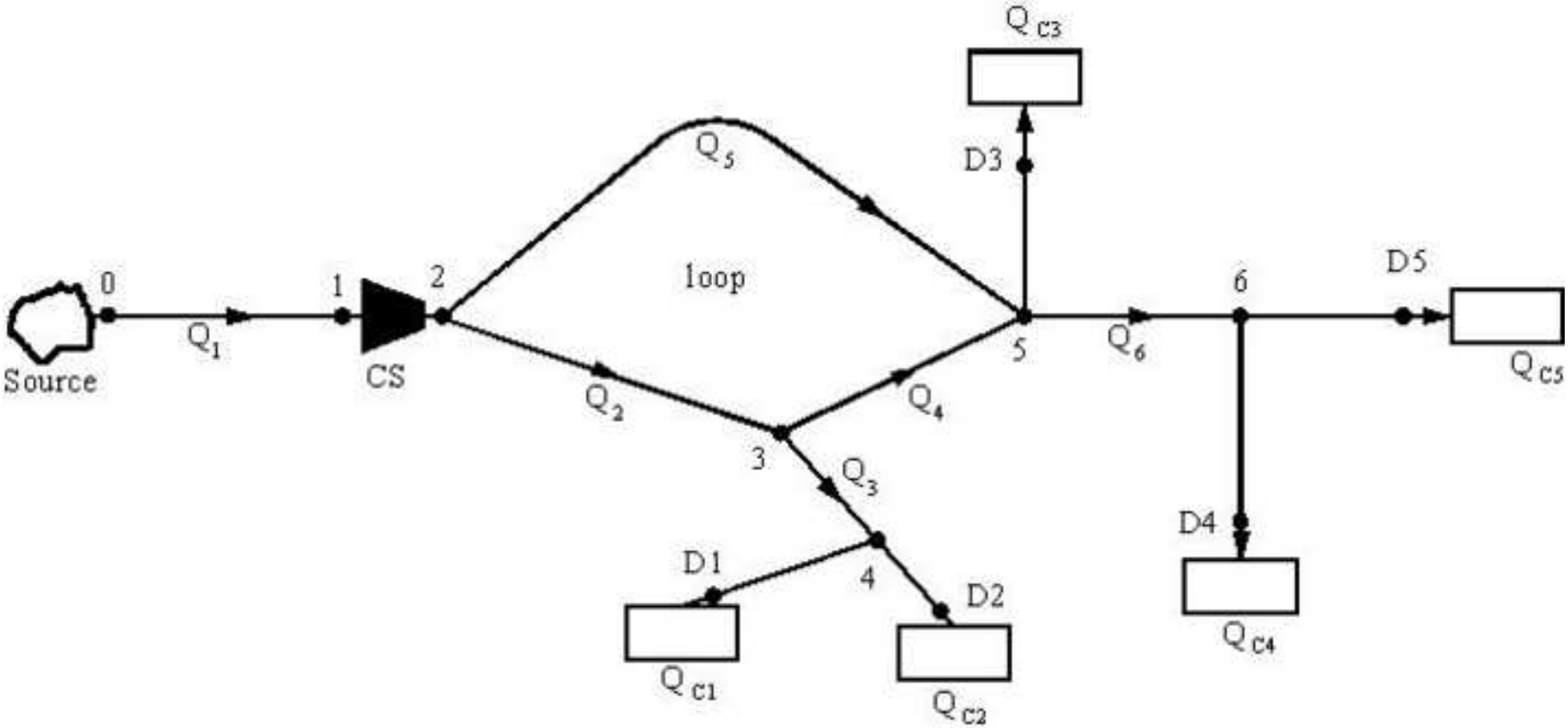

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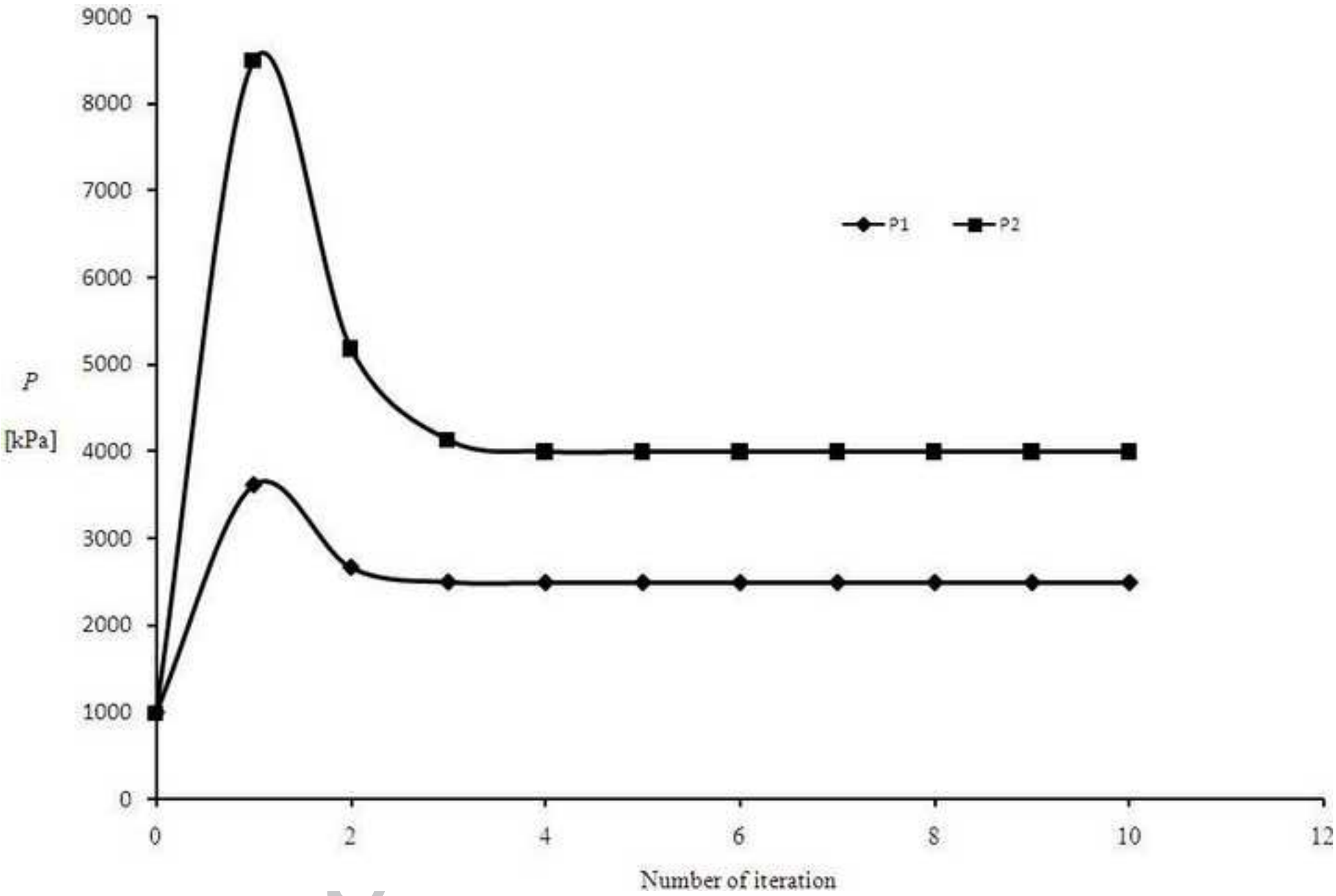

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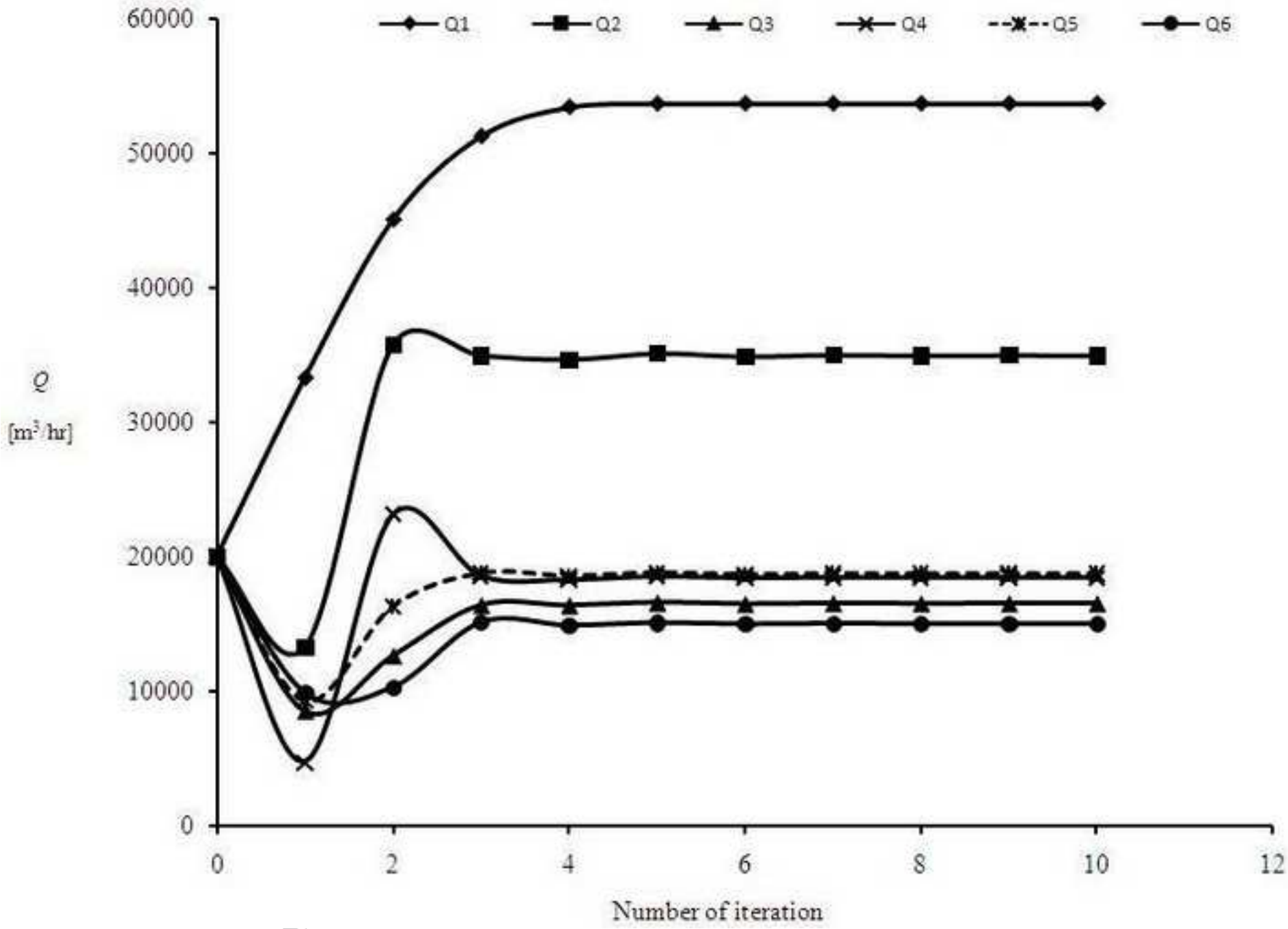

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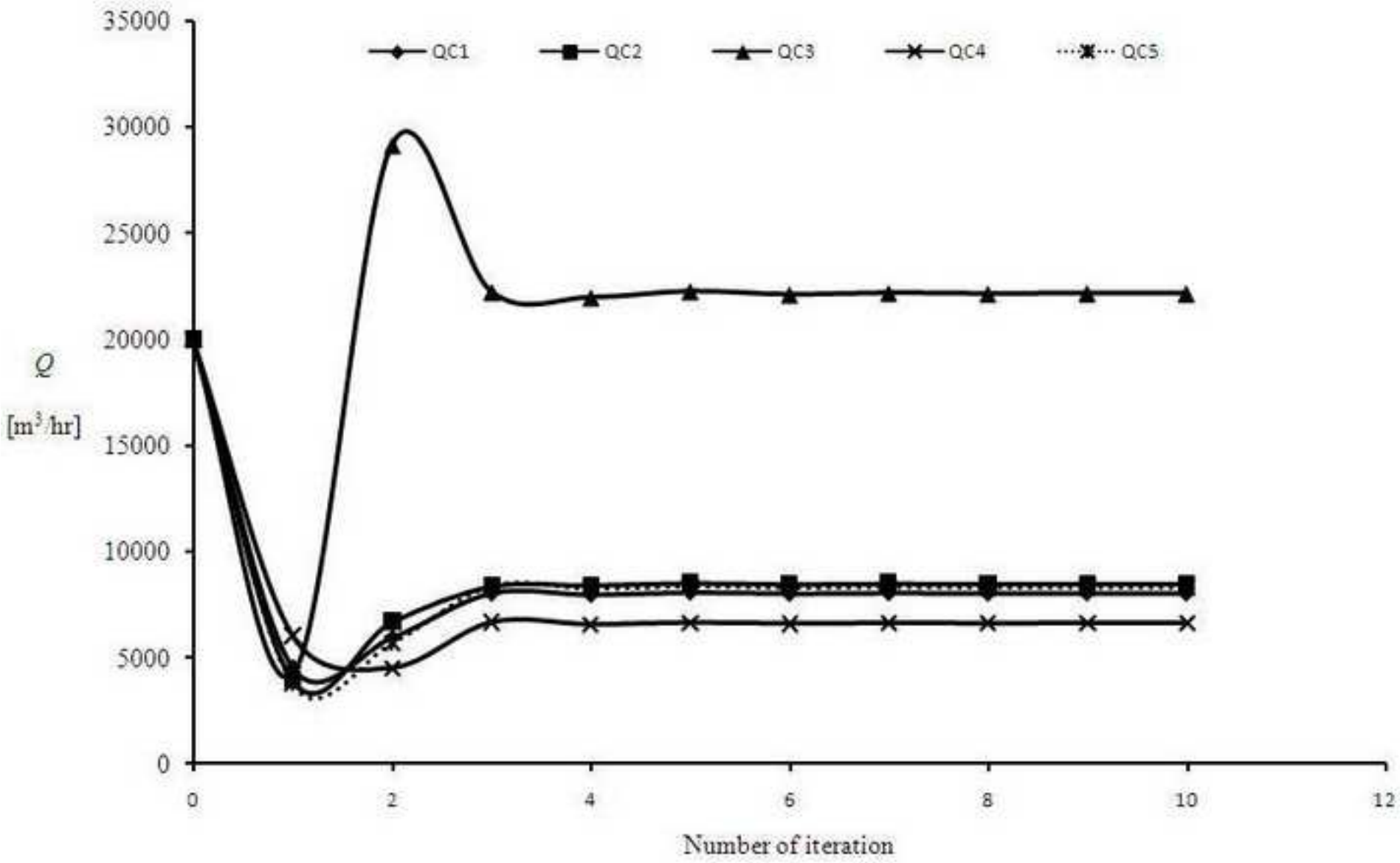

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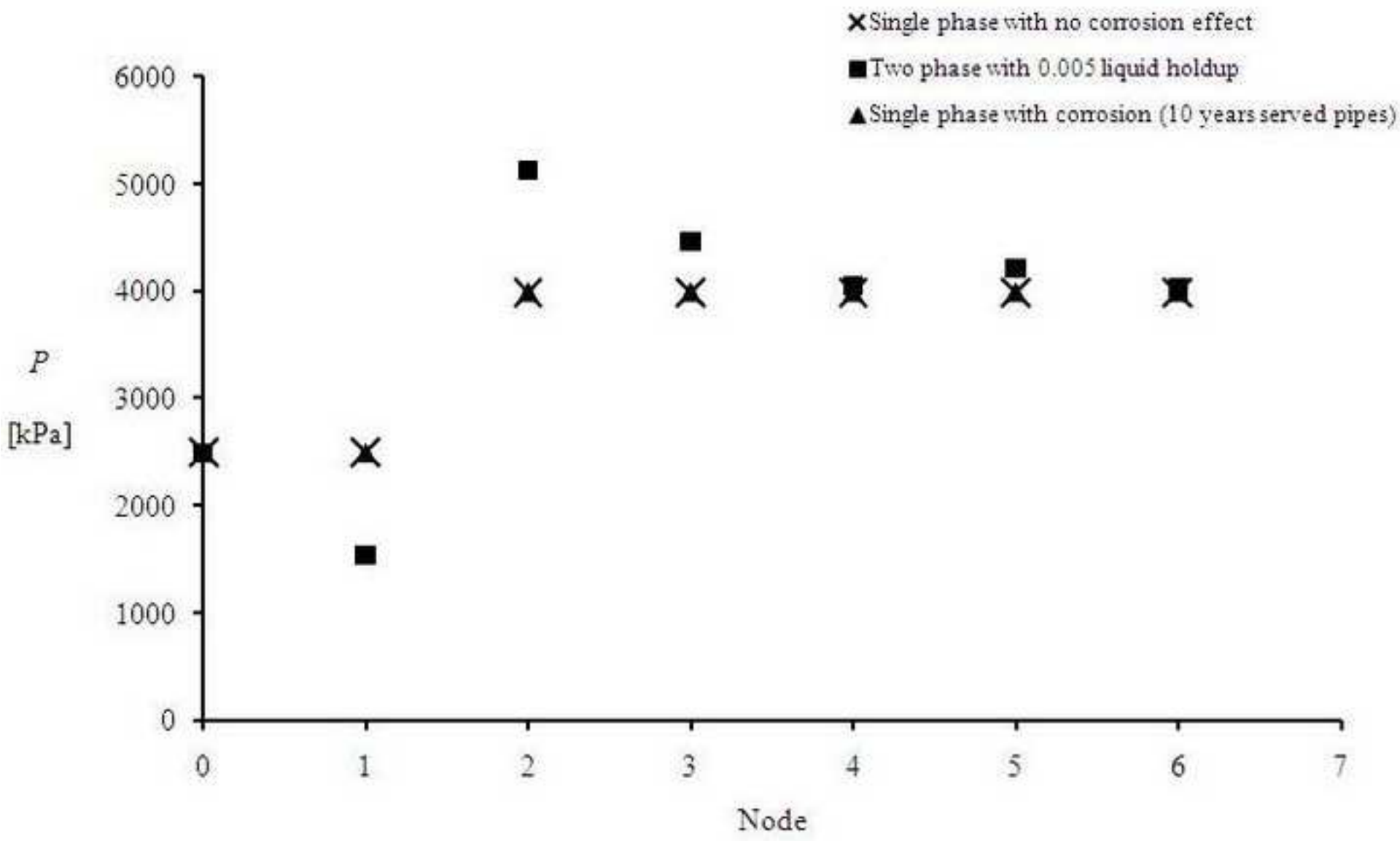

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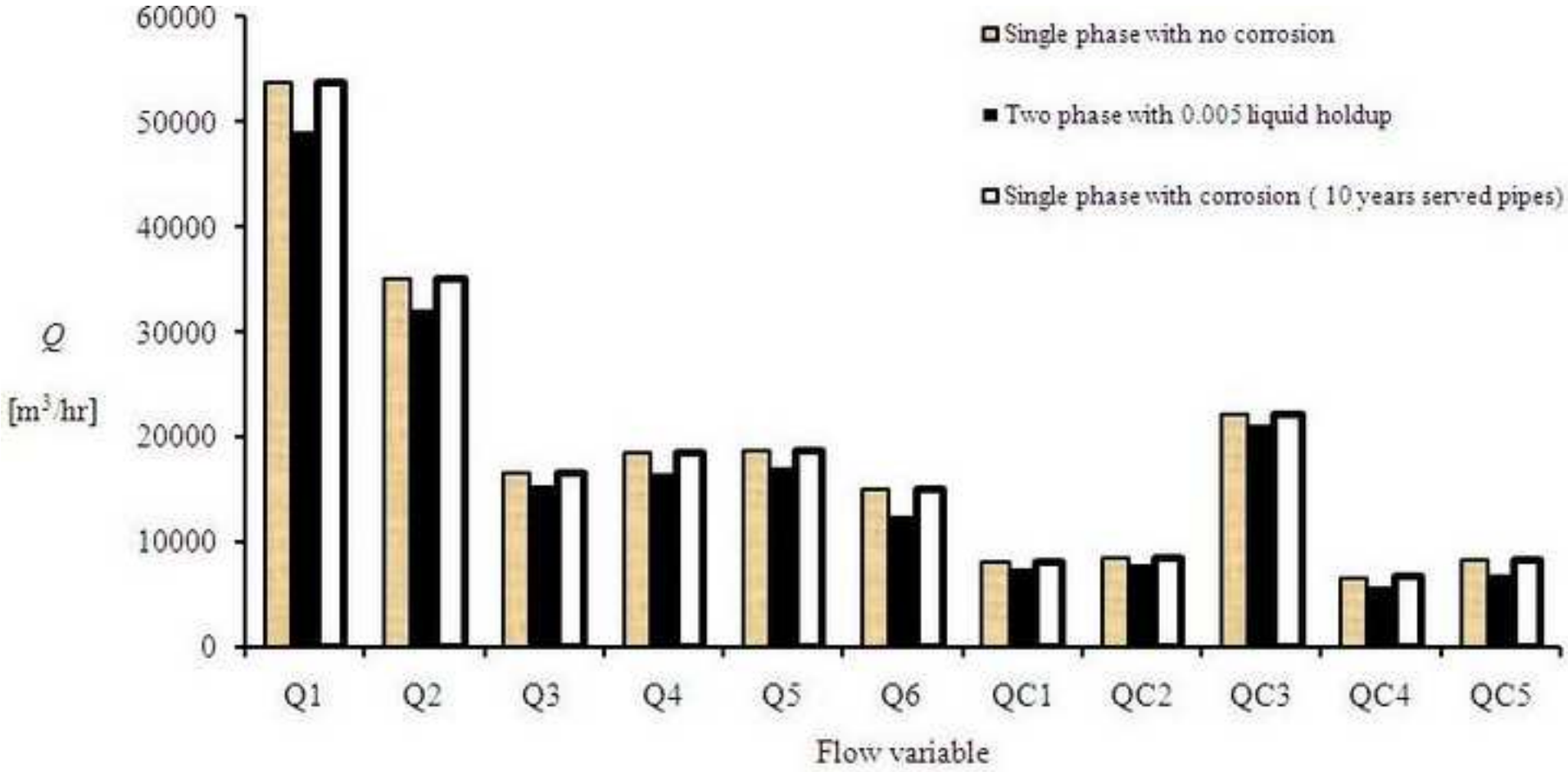

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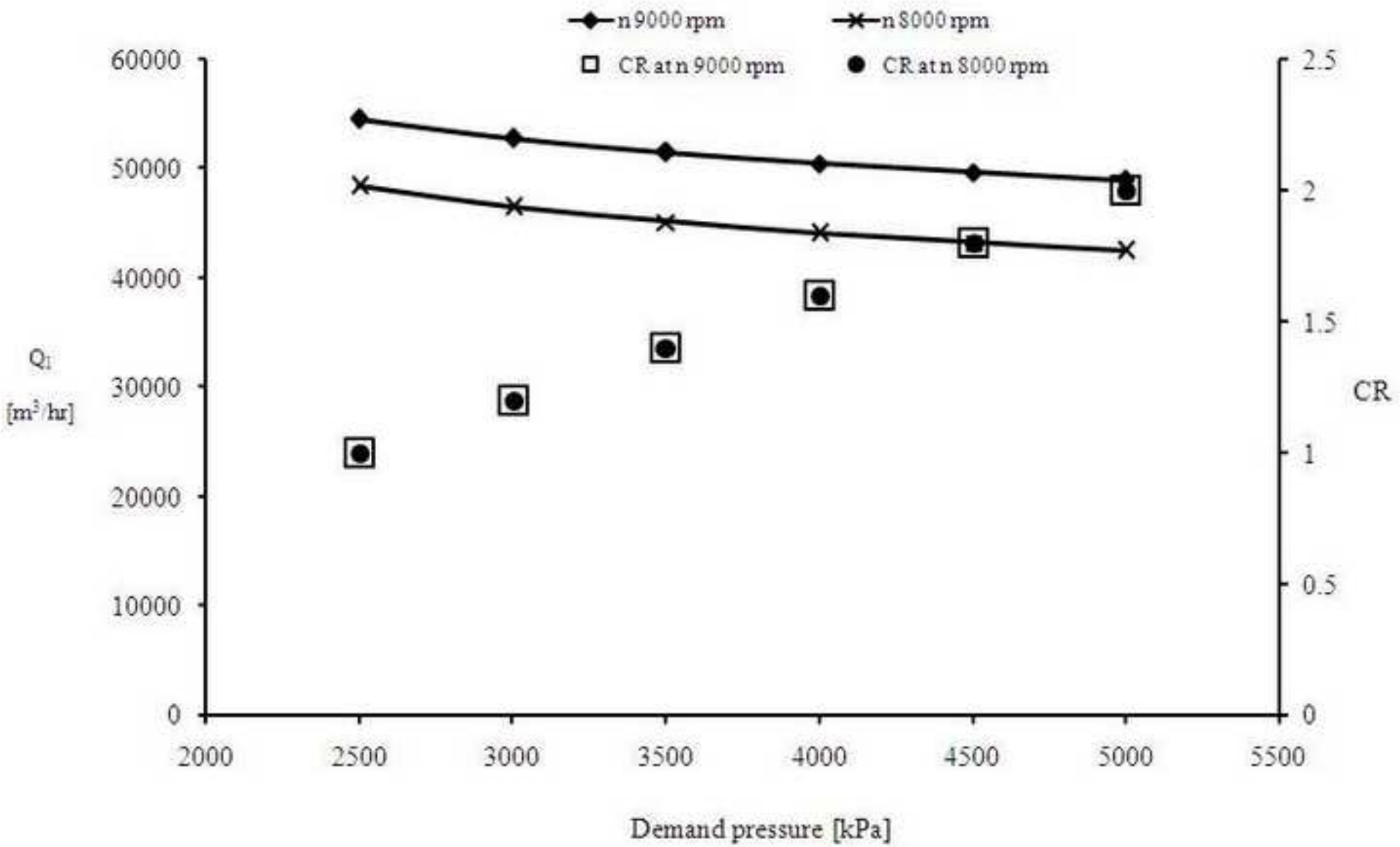

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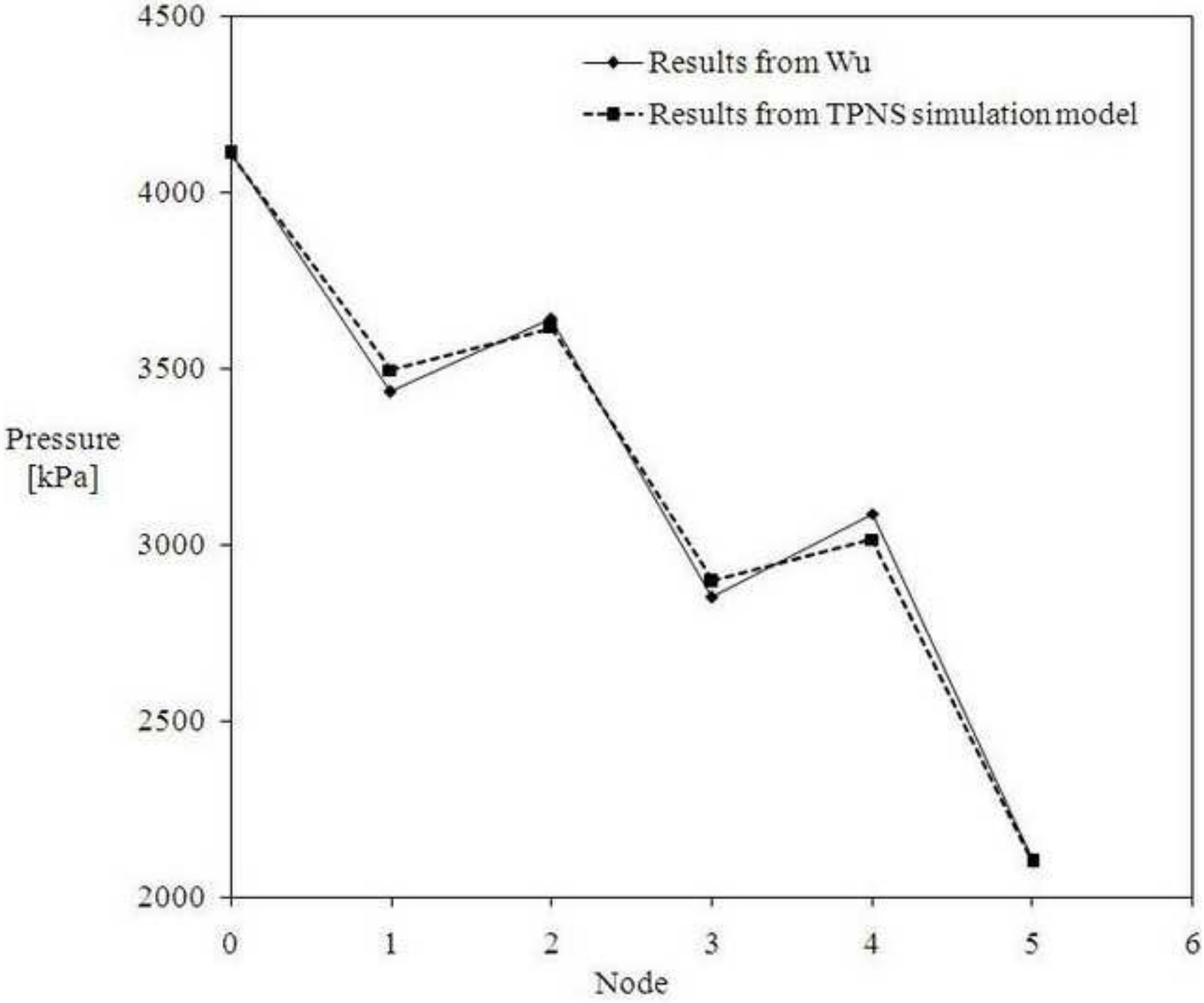

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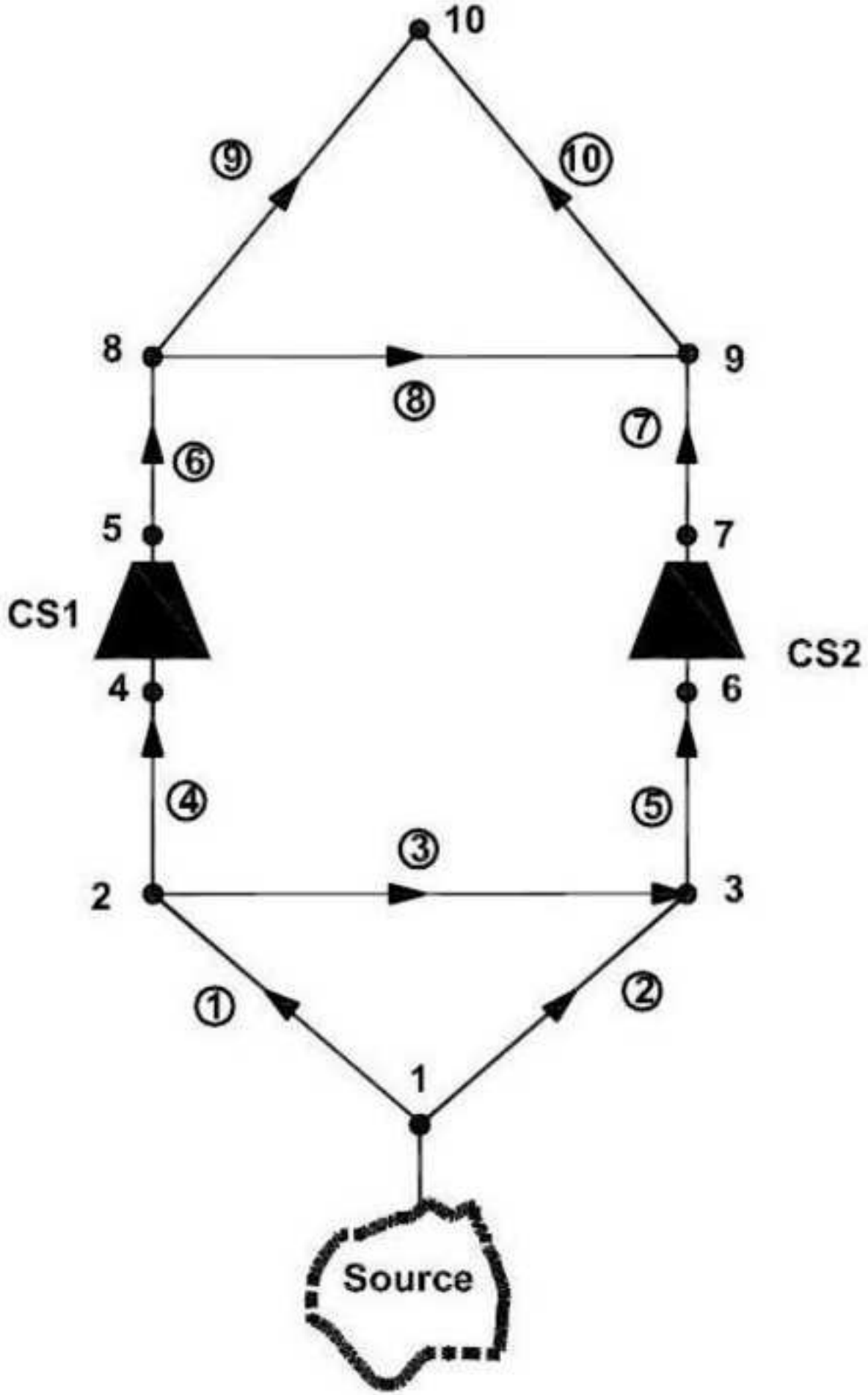

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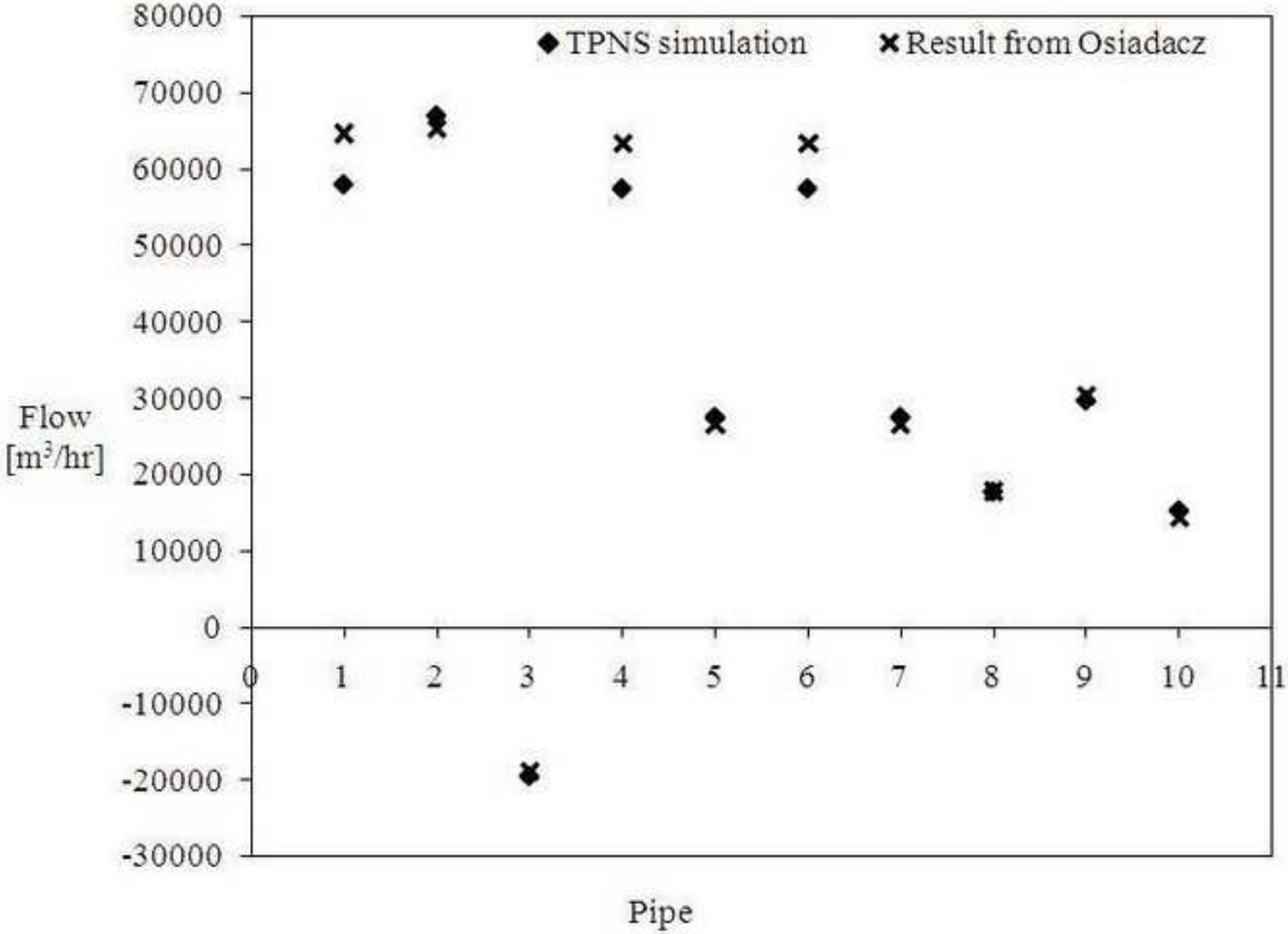

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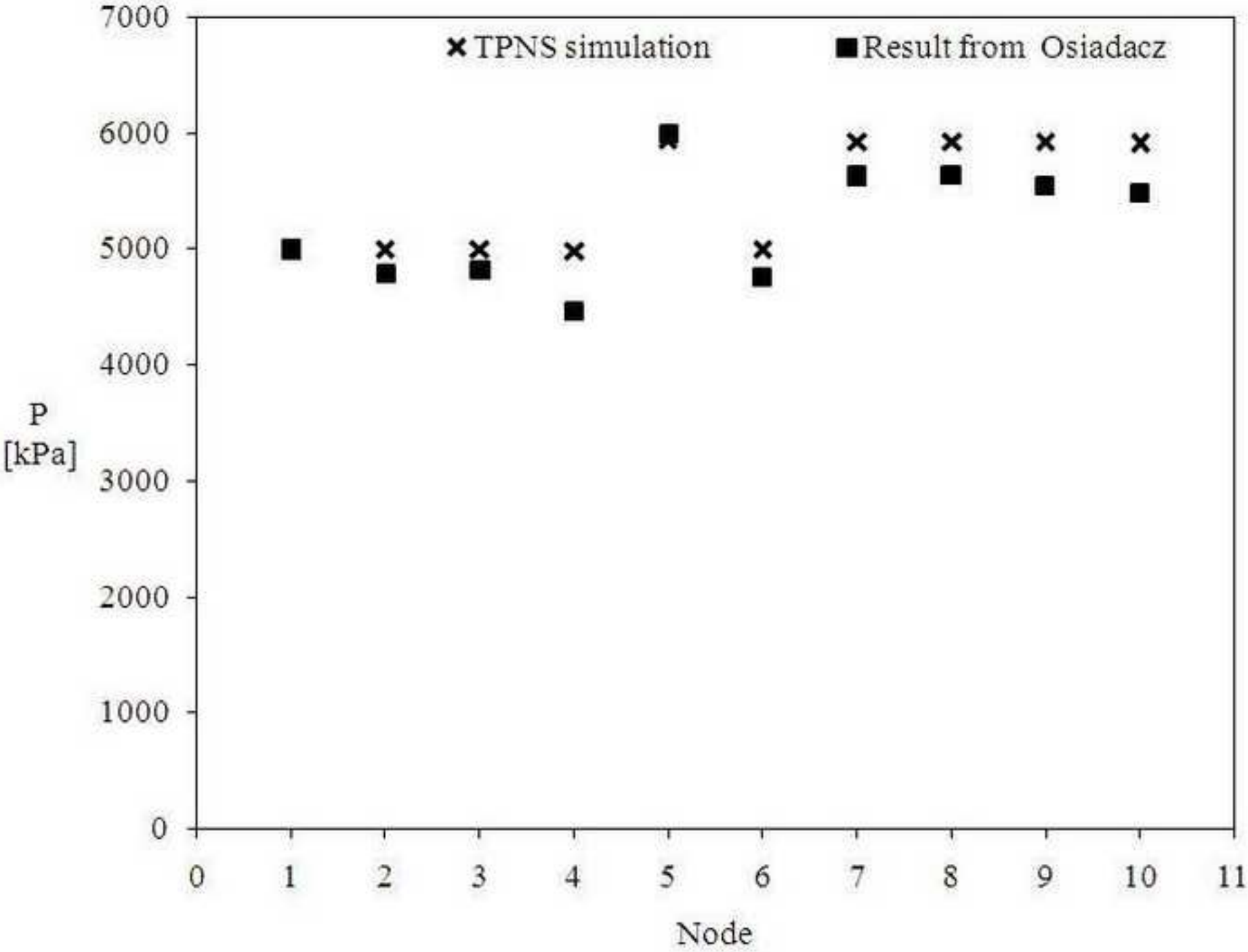
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